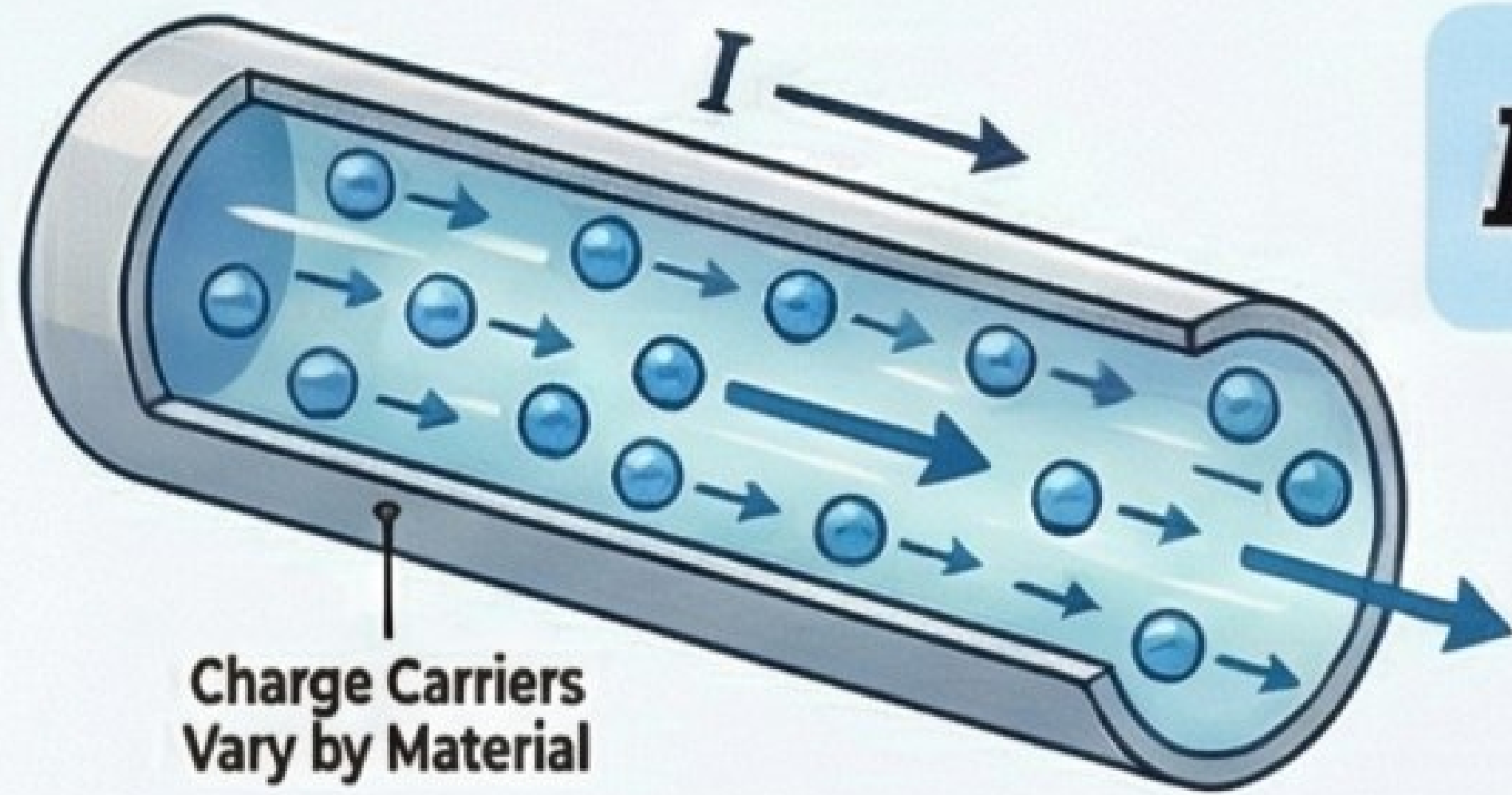


Current Electricity: A Visual Revision

Fundamentals of Electric Current



$$I = \frac{q}{t}$$

What is Electric Current?

Rate of flow of electric charge through a cross-section of a conductor.

Unit: Ampere (A) = Coulomb/second (C/s)

Current is a Scalar Quantity

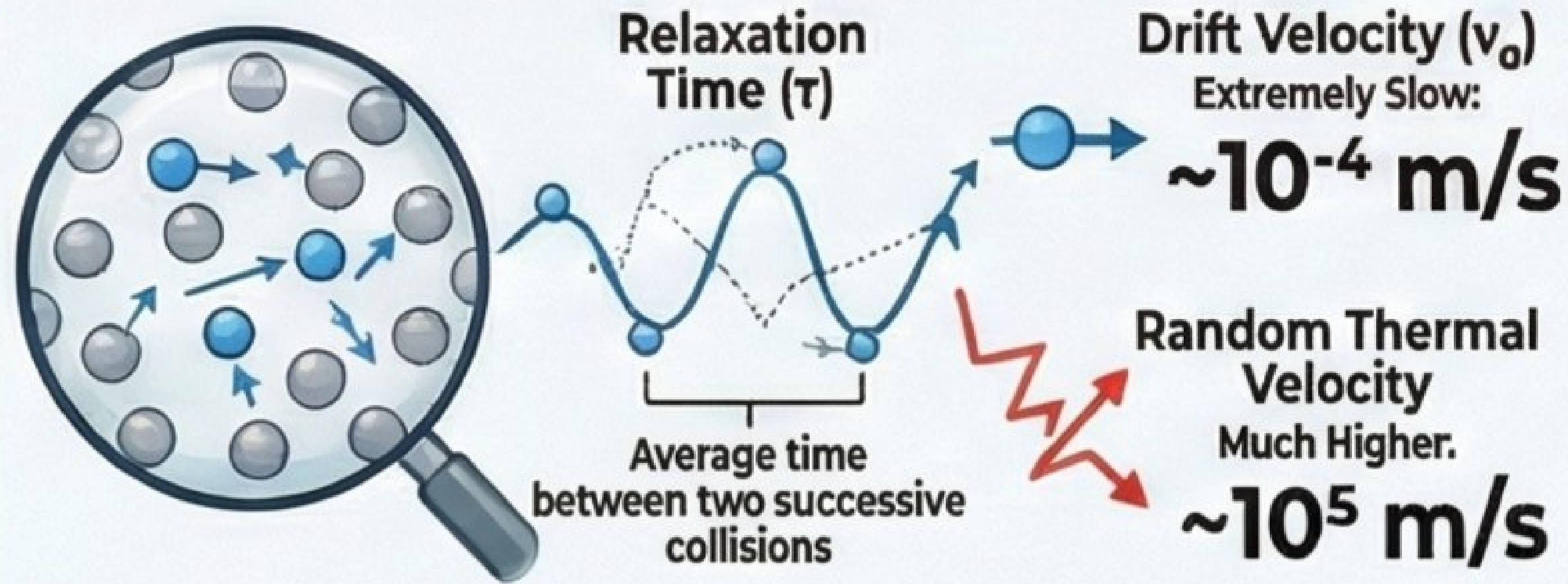
Adds algebraically, not vectorially.
Direction is direction of positive charge flow.

Current Density (J)

$$J = \frac{I}{A}$$



The Microscopic View of Conduction



Higher Temperature = Shorter Relaxation Time (τ)

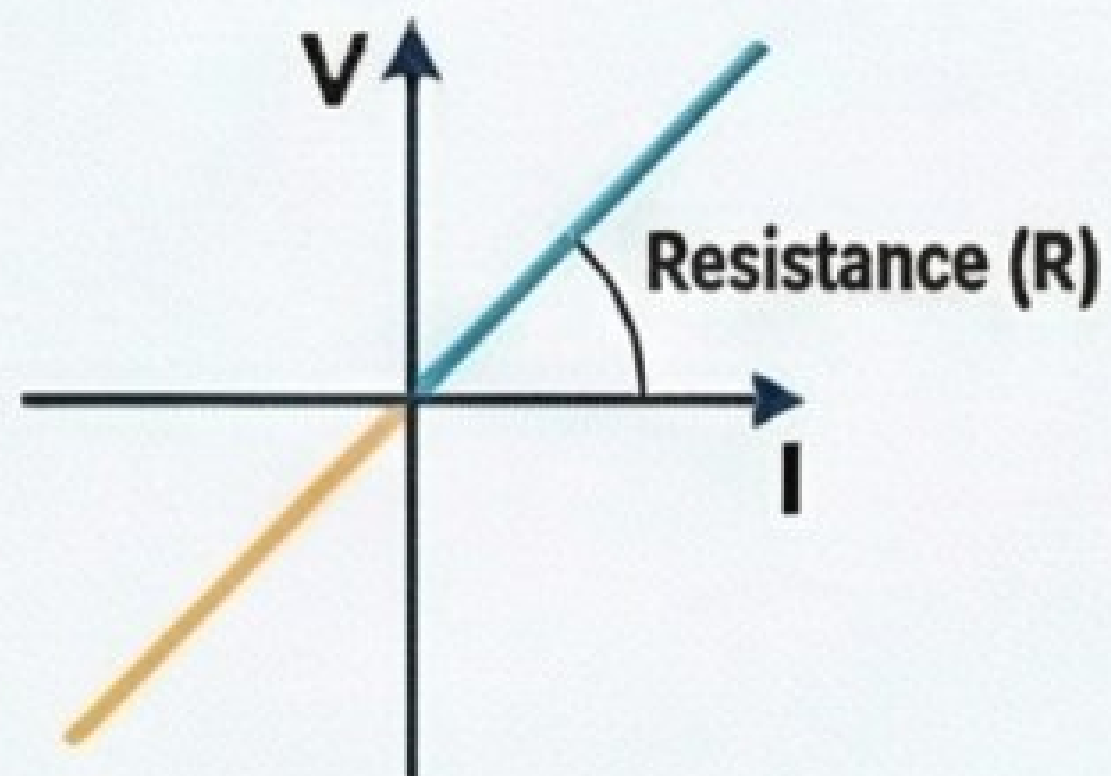
↑ Temp → ↑ Kinetic Energy
→ ↑ Collisions →
↓ Time between collisions.

$$\mu = \frac{v_d}{E}$$

Mobility (μ)
Measure of how quickly an electron can move.

Ohm's Law, Resistance & Resistivity

Ohm's Law
Current (I) is directly proportional to Potential Difference (V) (provided physical conditions constant).



$$V = IR$$

Resistance (R)
Obstruction to current flow from collisions.
Unit: Ohm (Ω)

Factors Affecting Resistance

$$R = \rho \left(\frac{L}{A} \right)$$

Directly proportional to Length (L)

Inversely proportional to Cross-sectional Area (A)

Stretching vs. Adding Length

Adding Length: Length doubled = Resistance doubled

Stretching: Length doubled = Resistance becomes FOUR times larger (n² times)

Resistivity (ρ)

Resistivity is a property of the material itself.

How Temperature Affects Resistance

Conductors (Metals)

↑ Temp →
→ ↑ Kinetic Energy →
→ ↑ Collisions →
→ ↓ Relaxation Time (τ)
→ ↑ Resistance.

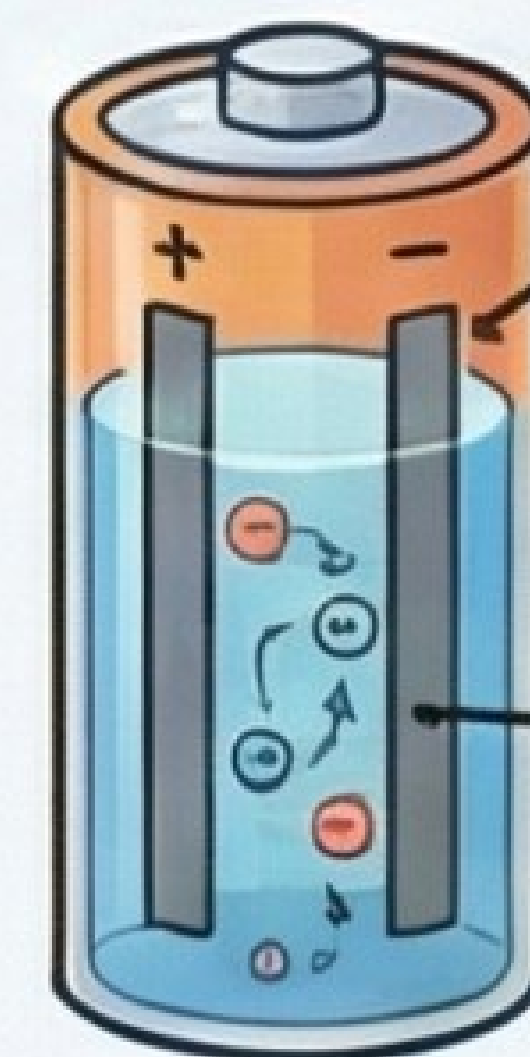
Alloys

Resistance increases slightly with temperature, but resistivity is very high.

Semiconductors & Electrolytes

↑ Temp →
→ Decreases resistivity and resistance
→ Better conductivity.

Cells: EMF & Internal Resistance



Electromotive Force (EMF or E)
Max potential difference when no current is drawn (open circuit)

Internal Resistance (r)
Resistance offered by electrolyte and electrodes to ion flow.

Factors Affecting Internal Resistance (r)

↑ Electrolyte Concentration
↑ Electrode Separation
↓ Temperature
↓ Larger Immersed Electrode Area

Discharging vs. Charging a Cell

Discharging:
EMF > Terminal Voltage (V)

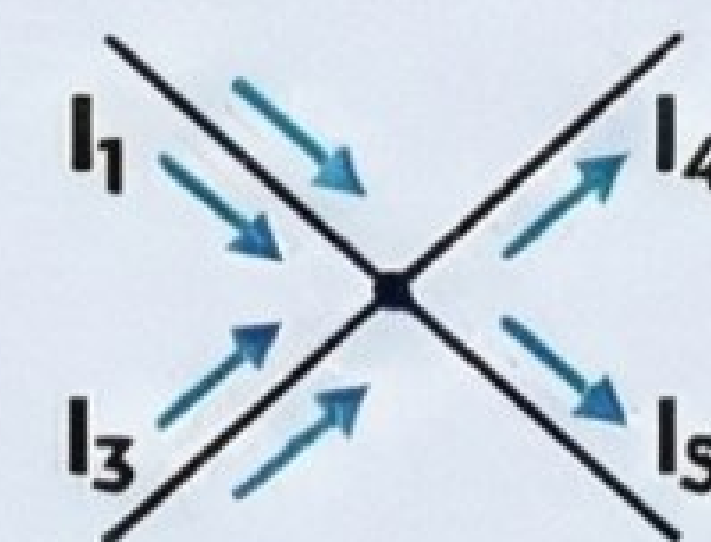
Charging:
Terminal Voltage (V) > EMF

Complete Circuit Current

$$I = \frac{E}{R + r}$$

Essential Circuit Analysis Tools

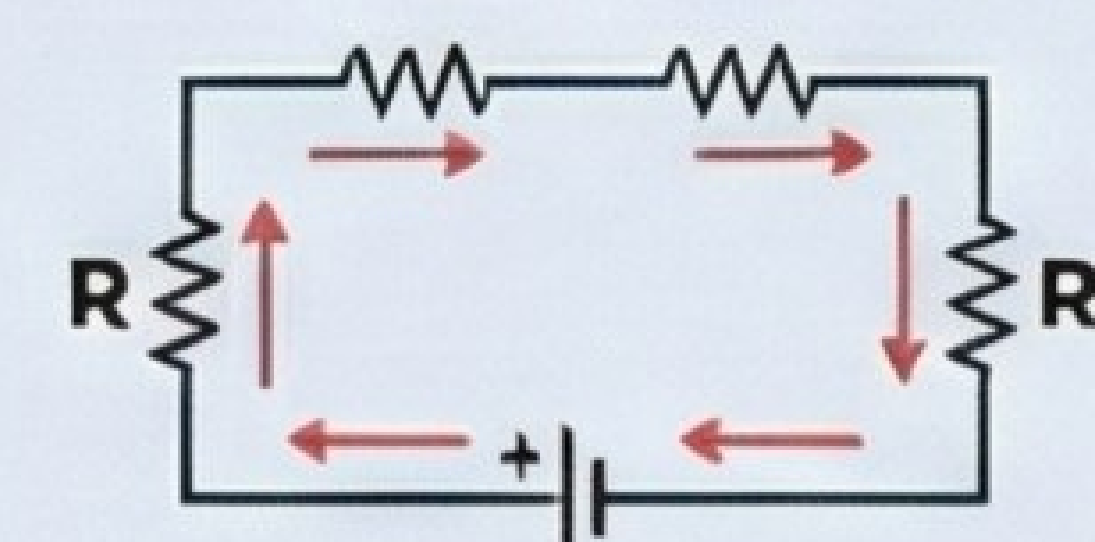
Kirchhoff's First Law (Junction Rule)



$$\sum I = 0$$

Algebraic sum of currents entering a junction is zero. Based on conservation of charge.

Kirchhoff's Second Law (Loop Rule)

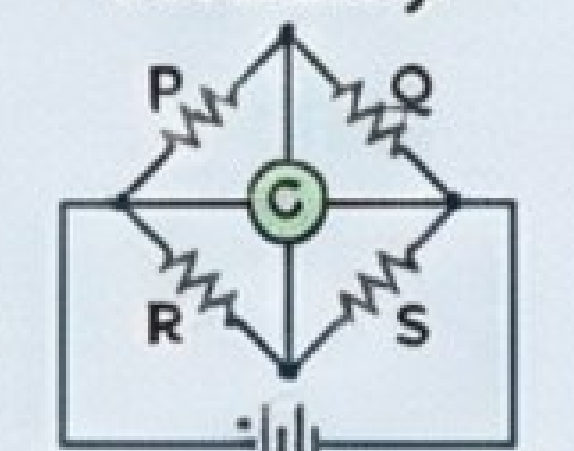


$$\sum \Delta V = 0$$

Algebraic sum of changes in potential around any closed circuit loop is zero. Based on conservation of energy.

Wheatstone Bridge

A circuit with four resistors used to measure an unknown resistance accurately.



Balanced Condition

The bridge is balanced when no current flows through the central galvanometer.